

Do Real Oil-Price Changes Still Predict Inflation? Evidence from G7 countries, 1992-2024

EC3030 Computational Economics

Spring Semester 2025-2026

Abstract

This project questions whether real-oil price changes still predict CPI inflation. To test this, the study uses annual data from the G7 countries. The dataset was obtained through 'Our World in Data' and 'World Bank'. Using the data, I estimated the country-level OLS regressions, pooled models, fixed-effects specifications, sub-period models, and lagged specifications. The results showed strong heterogeneity. US and Canada have the strongest oil-price relationship. France's relationship is found to be weak, and the relationship is found to be statistically insignificant elsewhere. The results also show that the pooled regression was weak before 2008 but has grown stronger after 2008. I also included a one-year lag model to consider the possibility of the effects of a change in oil price not being immediate, but over time. However, changes in oil prices due to oil supply and demand are not separately identified, therefore the results found should be interpreted as predictive associations rather than causal estimates.

Introduction

Blanchard-Galí (2007) argue that relationship has weakened, and as a result it has contributed towards inflation targeting becoming more realistic. They also argued that it has contributed to lower wage rigidity and a reduction in the intensity of oil-production. However, Kilian (2009) argued that the cause of change is important. Changes caused by OPEC cutting output is not the same as changes caused by global demand shocks. However, a single pooled coefficient treats them the same.

Two datasets are merged into a panel of seven countries starting in 1992 and ending in 2024. Annual CPI inflation measures the percentage change in CPI and is sourced from the World Bank's World Development Indicators. The oil price data is obtained from Our World in Data's real crude-oil series. The explanatory variable used is the year-on-year percentage change in real price.

A real-change series ensures the oil coefficient isn't picking up general inflation. Oil-price changes are weakly associated with inflation, and deflating to 2023 USD removes the mechanical effect of general price changes.

Table 1: Descriptive statistics

	Inflation	Inflation	Inflation	Inflation	Oil_Price_Change	Oil_Price_Change	Oil_Price_Change	Oil_Price_Change
	mean	std	min	max	mean	std	min	max
Country								
Canada	2.04	1.17	0.17	6.8	8.15	28.55	-47.06	69.49
France	1.67	1.12	0.04	5.22	8.15	28.55	-47.06	69.49
Germany	2.01	1.54	0.14	6.87	8.15	28.55	-47.06	69.49
Italy	2.42	1.85	-0.14	8.2	8.15	28.55	-47.06	69.49
Japan	0.49	1.14	-1.35	3.27	8.15	28.55	-47.06	69.49
United Kingdom	2.49	1.53	0.37	7.92	8.15	28.55	-47.06	69.49
United States	2.57	1.43	-0.36	8.0	8.15	28.55	-47.06	69.49

Table 1 shows mean annual CPI inflation peaking at 2.57% (US) and is lowest at 0.49% (Japan). The average oil-price change per year is 8.15% but the standard deviation is 28.55 which suggests high volatility. The final panel is composed of 33 observations and has no missing values.

Figure 2: Time series of inflation and oil prices

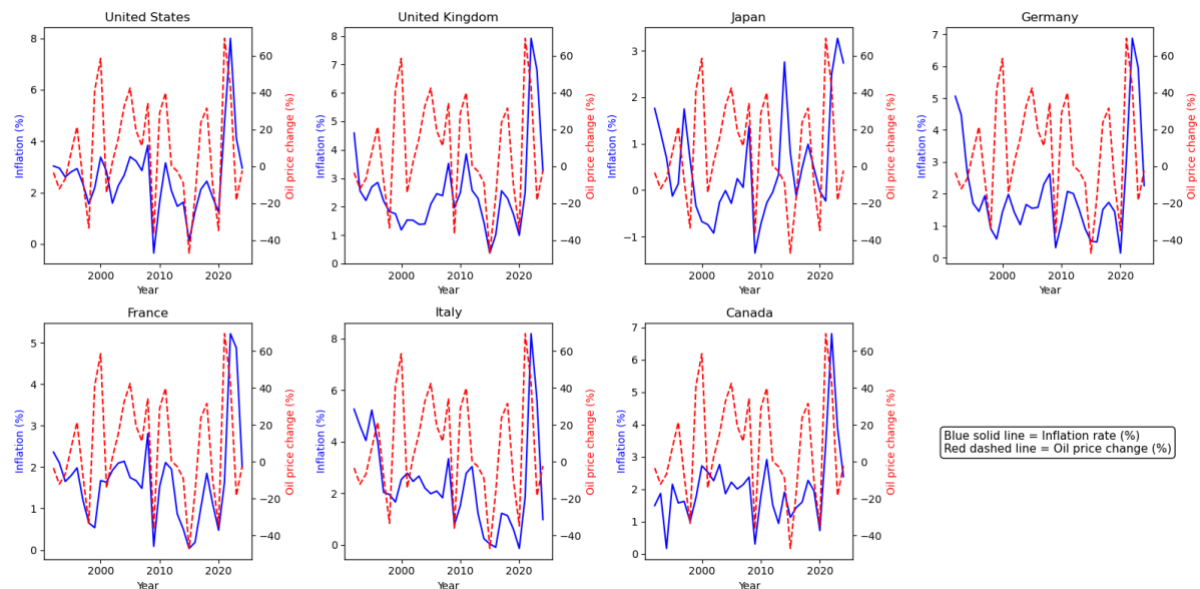


Figure 2 shows the relationship between inflation and oil price changes over time.

This project tests whether real oil-price changes have a positive relationship with CPI inflation. If and how the relationship differs across the G7 countries and if the relationship has changed after 2008. It also tests if oil-price changes affect inflation gradually.

Methodology

I used ordinary least squares regressions to estimate how strongly real oil-price changes are associated with consumer-price inflation.

Equation 1:

$$\pi_{i,t} = \alpha_i + \beta_i \Delta p^{\text{oil}}_t + \varepsilon_{i,t}$$

‘ $\pi_{i,t}$ ’ represents annual CPI inflation in country ‘i’. ‘ Δp^{oil}_t ’ is the year-on-year percentage change in the real crude oil price. The coefficient ‘ β_i ’ measures the percentage point change in CPI inflation associated with a one percentage point change in real oil price. The pooled regression treats the seven countries as one sample of 231 observations, giving an average response for G7 countries, at the cost of assuming every country shares the same slope.

The pooled observations are not independent because real oil-prices are common across countries within each year. Therefore, year-clustered standard errors for pooled and fixed effects specifications are also estimated. It therefore gives a more cautious inference because the oil-price variation comes from 33 annual observations rather than 231 fully independent observations.

To test whether the slope has changed across periods, I included a formal interaction term.

Equation 2:

$$\pi_{i,t} = \alpha_i + \beta_{i,0} \Delta p^{\text{oil}}_t + \beta_{i,1} \Delta p^{\text{oil}}_{t-1} + \varepsilon_{i,t}$$

Lags are built within each country’s own time series. Models are fitted in Python using the statsmodels package. The country-level regressions use conventional OLS standard errors. Pooled and fixed effects specifications are also checked using year clustered standard errors.

To test cross-country heterogeneity, oil-price changes are interacted with country dummies in a single pooled model. Two dynamic specifications add lagged inflation to control for persistence. The second also includes lagged oil-price changes. These help test whether the contemporaneous oil coefficient survives after accounting for last year’s inflation. An IV specification using lagged oil prices as an instrument was also considered. However, it was not continued as the lagged oil-price change has a direct effect on inflation, violating the exclusion restriction.

A limitation is that OLS captures the association, not the causation. Oil prices themselves respond to the same global demand conditions that drive inflation (Kilian,

2009). Therefore, the coefficient conflates supply and demand shocks. Monetary-policy stance, exchange rate, or labour market slack are also not explicitly controlled

Results

Figure 1: Oil price changes and CPI inflation by country, fitted OLS line

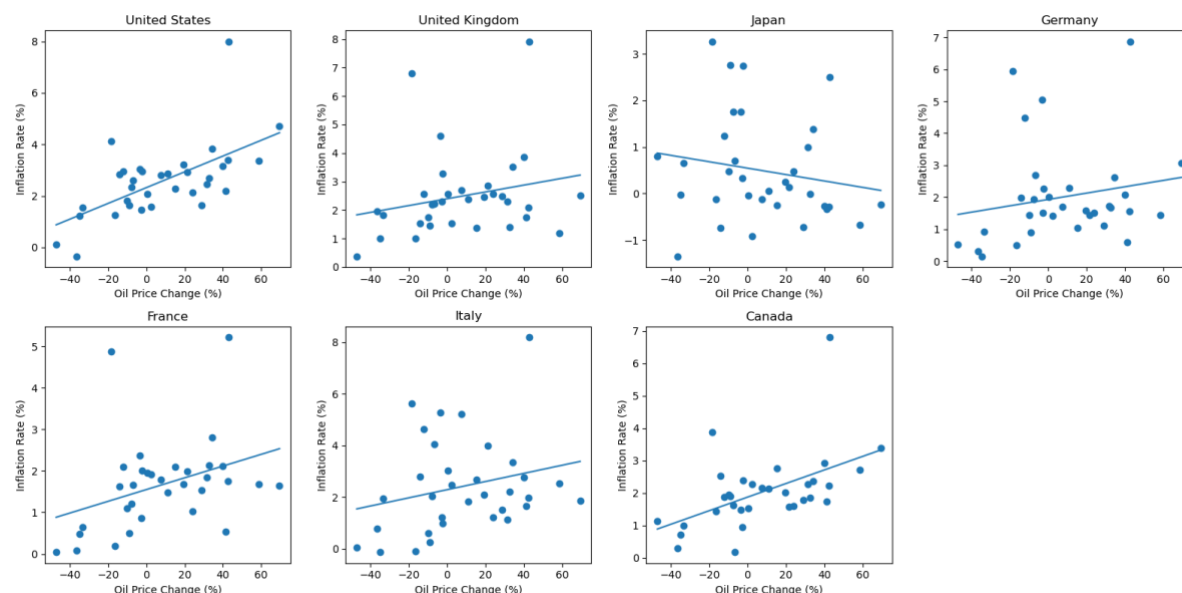
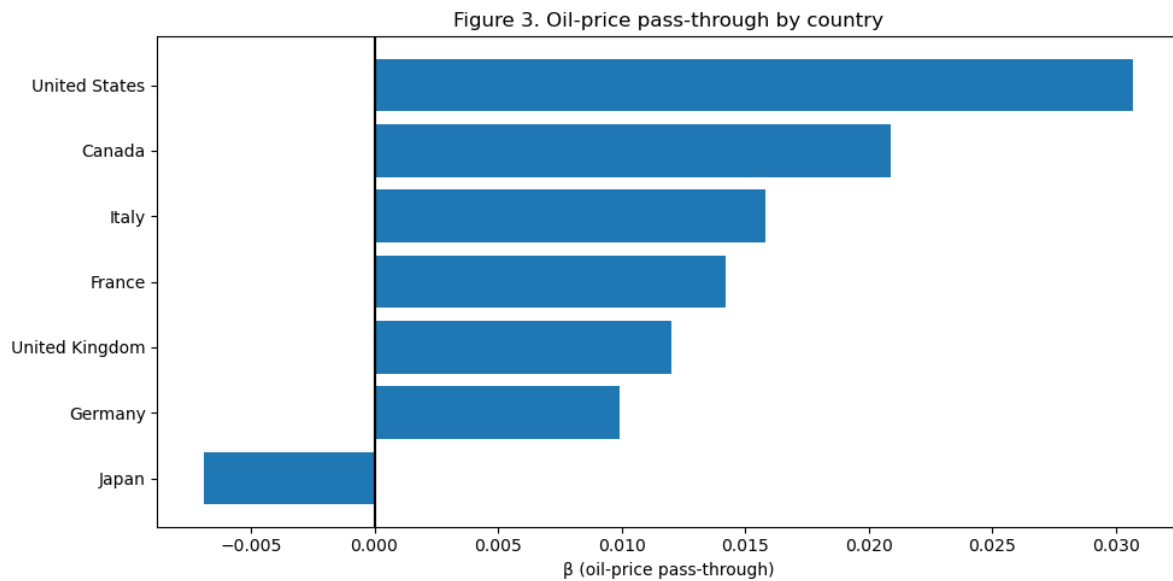


Table 2: Country-level OLS

Country	N	Beta	Std Error	T Stat	P Value	Sig	R Squared
United States	33	0.0307	0.0071	4.32	0.0001	***	0.376
Canada	33	0.0209	0.0063	3.32	0.0023	***	0.263
Italy	33	0.0158	0.0113	1.4	0.1725		0.059
France	33	0.0142	0.0066	2.16	0.0385	**	0.131
United Kingdom	33	0.012	0.0094	1.28	0.21		0.05
Germany	33	0.0099	0.0095	1.04	0.3063		0.034
Japan	33	-0.0069	0.0071	-0.97	0.338		0.03

The table shows that in the same year 100 percentage-point increase in real oil-price is roughly associated with a 3.07 percentage point increase in the US's CPI inflation. The scale of the increase is consistent with Choi et al. (2018). The table shows that the pass-through is strongest and most precisely estimated in the US and Canada. The table also shows that France's coefficient is significant at a 5% level ($\beta = 0.014$, $p = 0.039$) and UK, Germany, and Italy are statistically insignificant. On the other hand, Japan's point estimate is negative. This questions if the G7 can be treated together and if the differences between countries are too different to pool together.

Figure 3: Coefficient bar chart



Pooled OLS returns are $\beta = 0.014$ ($R^2 = 0.06$) and the country fixed effects produce the same coefficient. The within R^2 of 0.077 is the better fit measure for fixed effects. The common intercept is statistically rejected as the F test for poolability shows a highly significant result $F(6,223) = 9.21$, $p < 0.001$). Each G7 country have structural differences such as different inflation targets, central bank credibility and openness to trade resulting in each country having a different baseline inflation rate. Pooling estimates therefore hides the variation between countries shown in Figure 3. These results support Kilian (2009) and Choi et al. (2018) as they warn against interpreting pass-through estimates without accounting for country controls. Beyond considering variation between countries, the oil-price relationship may have shifted over time.

Table 4: Summary of all model specifications

Specification	Beta	Std_Error	P_Value	R_Squared	N
Pooled OLS	0.0138	0.0079	0.0806	0.063	231
Panel FE (country)	0.0138	0.0079	0.082	0.077	231
Pre-2008 (pooled)	-0.0049	0.0047	0.3024	0.01	112
Post-2008 (pooled)	0.024	0.0049	0.0	0.171	119

The pooled OLS has a positive coefficient but is only marginally significant once year-clustered errors are included ($\beta = 0.0138$, $p = 0.081$). Post-2008, the results show that there is a clear change. Pre-2008 the pooled coefficient is negative and statistically insignificant ($\beta = -0.0049$, $p = 0.302$). Post-2008 it becomes positive and statistically significant ($\beta = 0.0240$, $p < 0.001$), suggesting that oil-inflation relationship has strengthened as time has gone on.

Table 6: Robustness

Specification	Beta	Std Error	P Value	R Squared	N
Post-2008 (full)	0.024	0.0049	0.0	0.171	119
Post-2008 excl. 2022	0.0147	0.0041	0.0005	0.105	112
Post-2008 excl. 2020 and 2022	0.0128	0.0044	0.0044	0.076	105

Table 6 shows that the results survive the robustness check which suggests that post-2008 increase is genuine. This contrasts with Blanchard-Gali (2007) declining-pass-through hypothesis. One likely reason for this is due to composition of shocks changing. With events from the post 2008 oil movements to the 2022 Russia Ukraine conflict now being included. The effects of these shocks most likely transmitted to consumers differently than the OPEC episodes did. Another reason could be that zero-lower bound policy rates have worn out central banks' capacity to offset cost-push shocks. Another possible reason is as supply chains have now been globalised, it has led to leaner inventories causing downstream transmission to increase (Di Giovanni et al., 2023).

The country level data helps distinguish which channel here matters most. As Japan's post 2008 coefficient has been flat ($\beta = 0.000$). It is difficult to only assume it has been a product of their zero lower bound monetary policy as they have experienced it longer than any other G7 country, yet their oil-price relationship did not improve. Meanwhile, countries like the US where oil production intensity is high and consumer pass-through is high saw an increase. This suggests that supply-chain explanation may be more accurate than the monetary-policy constraint despite them not being mutually exclusive. It also suggests that if pass-through does primarily work through supply chains, it should therefore show the effect accumulating over time rather than instant.

Table 3: Baseline vs lagged-model comparison

Country	R2 Baseline	R2 Lagged	Improvement	Lag Coef	Lag P
United States	0.384	0.517	0.132	0.0183	0.0086
United Kingdom	0.063	0.229	0.167	0.0212	0.0182
Japan	0.026	0.047	0.021	0.0056	0.4344
Germany	0.05	0.212	0.162	0.0204	0.0211
France	0.139	0.366	0.227	0.0186	0.0031
Italy	0.075	0.247	0.172	0.026	0.0155
Canada	0.26	0.416	0.157	0.0162	0.0092

Table 3 shows that once the first lag is added, it raises the R^2 in every G7 country apart from Japan. This suggests that the hypothesis was correct, and that consumer prices respond to oil-price changes gradually.

Table 5: Cumulative pass-through

Country	Beta Year 0	Beta Year 1	Cumulative
United States	0.0298	0.0183	0.0481
Italy	0.0154	0.026	0.0413
Canada	0.0196	0.0162	0.0358
United Kingdom	0.0115	0.0212	0.0327
France	0.0133	0.0186	0.0319
Germany	0.0099	0.0204	0.0304
Japan	-0.0067	0.0056	-0.0011

Table 5 shows that the US has the largest cumulative pass-through.

Figure 4: Baseline vs lagged R^2

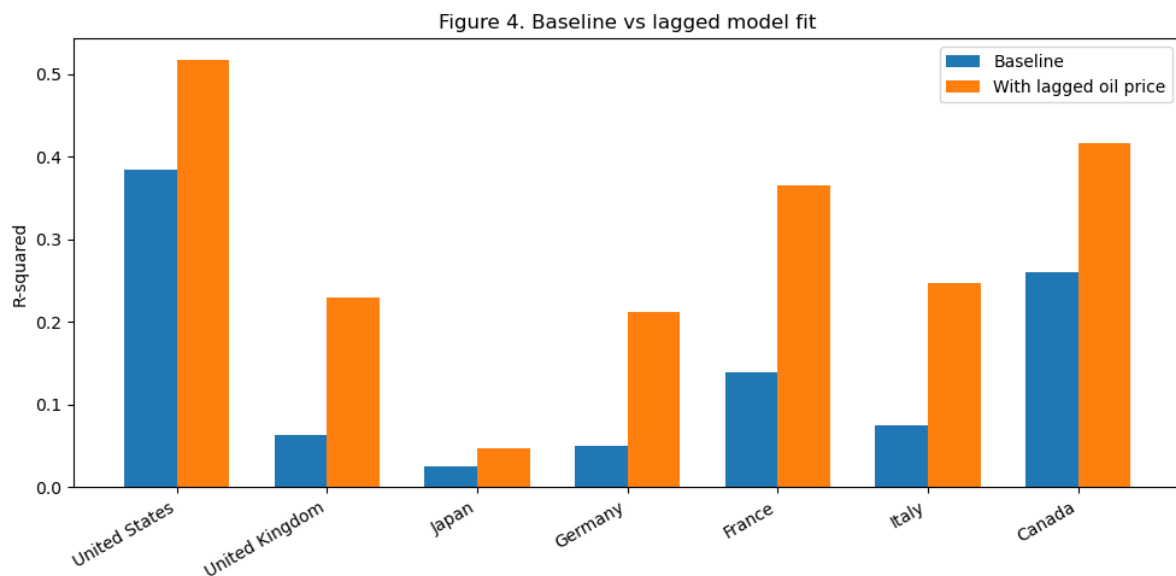
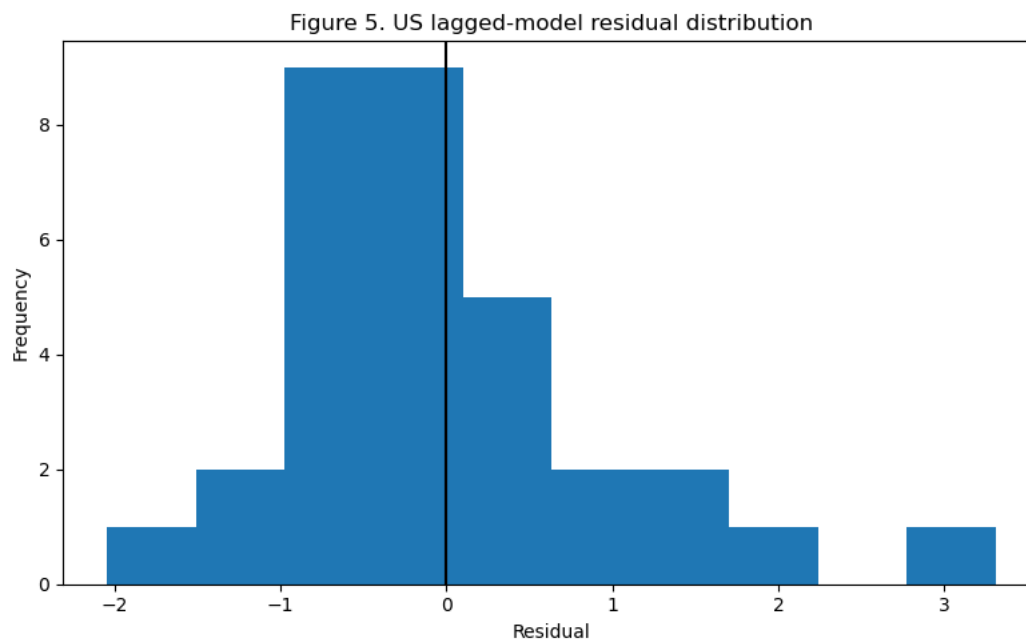


Figure 4 shows the comparison between R^2 of the baseline and lagged specifications across countries.

Figure 5: US lagged model residual distribution



The graph shows the residual distribution from the US lagged model is approximately symmetric around 0, showing no obvious outliers. This supports the standard OLS error assumptions.

Discussion

Japan is the clearest outlier. Their full-sample coefficient is negative and insignificant, and its post-2008 coefficient is close to zero. This may reflect Japan's prolonged low-inflation environment, weaker inflation expectations, energy-efficiency improvements, or differences in monetary and exchange-rate transmission. However, these factors are not directly tested in this project. Therefore, Japan should be interpreted as evidence of cross-country heterogeneity rather than proof of an explanation.

Limitations

There are several limitations worth flagging. OLS only captures association, not causation. Oil prices respond to the same global demand conditions that drive inflation, so the contemporaneous coefficient mixes supply-driven and demand-driven shocks together (Kilian, 2009). Also, the annual data hides within-year dynamics. Using monthly data would capture the transmission lags that prior studies typically find at 3-9 months. Only 33 observations limits the statistical power and means that insignificant results for the UK, Germany, and Italy cannot be interpreted as evidence of no relationship. It can only be viewed as no relationship was found within this data. Durbin-Watson statistics of 0.77-1.0 across the pooled and country regressions indicate positive correlation in the residuals. Therefore, the reported standard errors likely understate true uncertainty. Year-clustered or HAC-robust errors would provide a more conservative inference.

Monetary policy stance, exchange rate movements, and labour market slack are not explicitly controlled for.

A useful next step would be a structural VAR following Kilian (2009) to separately identify supply and demand shocks, alongside heteroskedasticity-robust standard errors and monthly data. Another concern is that the CPI inflation is the dependent variable. Therefore, the effect of energy prices in the consumer basket are included. A useful solution would be to compare the results with core inflation, which excludes energy and food prices.

Policy implications

These results contrast with pre-2008 view that oil had been decoupled from advanced-economy inflation. Inflation forecasts that downplayed energy inputs in the 2010s may need to be re-evaluated given the oil-price relationship has strengthened since 2008.

Conclusion

Overall, there is evidence to suggest that the real oil-price changes still predict CPI inflation, but the relationship being uneven over time and countries. The strongest evidence is for the US and Canada, while Japan shows little pass-through. The pooled relationship is weak before 2008 but stronger after 2008. Adding a one-year lag improves explanatory power in most countries, suggesting that part of the association operates gradually. However, because the analysis uses OLS and does not structurally identify oil supply and demand shocks, the findings should be interpreted as predictive associations rather than causal estimates.

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